

TransUnion Client Integration Document

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Introduction

This document and accompanying materials will help you quickly integrate with the Valifi APIs. It includes:

1. Postman Collection
2. Typical Customer Journey and Flow
3. Authentication
4. Address Lookup
5. API Credit Report
6. Summary Report
7. Batch Data
8. PDF as a Stream

9. Testing
10. Identifying Elements in the Report

Postman Collection

A Postman collection will be provided alongside this integration document to help you interact with the Valifi platform UAT environment.

Modify the collection variables with your provided username and password.

Typical Customer Journey and Flow with OTP

Customer Journey

1. Visits your landing page
2. Enters name, address, contact details and date of birth
3. Sent 6 digit code to mobile phone
4. Enters code into your application form
5. Mobile and AML checked with Valifi
6. Presented with a list of HP agreements
7. Capture e-sign etc.
8. Alert 24 hours later with any additional agreements found

Application Flow

1. Enters name, address, contact details and date of birth
2. Pass your credentials securely to the Valifi authentication endpoint
3. Retrieve an access token that will be used with every subsequent request
4. Optionally pass the postcode to Valifi API for list of matched addresses
5. Allow customer to pick the correct address
6. Pass customers mobile to Valifi OTP service

7. Valifi sends customer 6 digit pass code
8. Customer enters pass code into your application
9. Pass mobile and passcode back to Valifi
10. Valifi returns pass / fail response
11. Pass customer details to Valifi Mobile ID / AML service
12. Valifi returns AML and Mobile confidence indicators
13. You decide if customer has been sufficiently verified
14. Pass the same address to Trans Union and / or Equifax search endpoints
15. Interrogate the raw credit file or summary, store the results and optionally return information back to the customer
16. Call the batch API or request a previous search 24 hours later to retrieve any historic finance agreements that were not found in the standard CZ credit report

Notes:

- The token will expire in 60 minutes and will need to be reauthenticated. It is recommended that you catch any 403 errors and reauthenticate as required. This would typically be with an Axios interceptor in a React platform.
- The TransUnion and Equifax endpoints can return a full credit report, a summary of motor finance agreements, a PDF summary of the CZ report or a PDF of the account summary.
- In most cases the summary report will be the easiest data to work with, showing a clear array of motor finance agreements, without having to crawl the CZ report.

Authentication

The authentication method is common across all Valifi APIs.

To access data APIs, you must first obtain an access token. This token will automatically be stored in the collection variables. You are responsible for keeping your user credentials and access token secure.

In a production environment, real users must verify their identity with a 2FA SMS code. API users must submit their calling IP addresses for whitelisting. Access tokens expire after 8 hours, at which point you must request a new one.

Post: <https://staging-app.valifi.click/basic-auth>

Request

Your supplied client ID (username) and secret (password) should be sent in the header as Basic Auth

Response

```
{
  "status": true,
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJsInVz
  }
}
```

Address Lookup

The address lookup service allows you to search for addresses using a UK postcode. The addresses returned will be in an optimal format to pass into the Equifax CZ or KBA search request, increasing the chances of a customer match.

In addition to the formatted address, each returned address will contain a unique address identifier. This address identifier can be passed into the Equifax CZ and KBA search, instead of the full address. The unique address identifier is NOT accepted by TransUnion.

Post: <https://staging-app.valifi.click/bureau/v1/equifax/postcode-lookup>

Request

```
{
  "clientReference": "test",
  "postCode" : "BA13 3BN"
}
```

Response

```
{
  "success": true,
  "data": {
    "listAddressByPostcodeResponse": {
      "clientRef": "test",
      "totalResults": 869,
      "matchedStructuredAddress": [
        {
          "county": "WILTS",
          "name": "BLENLEY LODGE",
          "postcode": "BA13 3BN",
          "postTown": "WESTBURY",
          "street1": "HIGH ST",
          "addressID": 58150004506
        },
        {
          "county": "WILTS",
          "name": "BRIARFIELD",
          "postcode": "BA13 3BN",
          "postTown": "WESTBURY",
          "street1": "HIGH ST",
          "addressID": 58150004256
        }
      ]
    }
  }
}
```

TransUnion Credit Report

A credit search can be performed by passing in a customer's name, date of birth, and current address.

It is your responsibility to ensure you have completed appropriate KYC checks and obtained explicit consent from your customers before accessing credit data.

Note: Including the PDF in the response can significantly increase overhead and response time. We recommend retrieving the PDF separately through the dedicated PDF endpoint after completing the search.



When testing TransUnion reports, use the surname "AutomotiveFinance" to retrieve sample motor finance records

Post: <https://staging-app.valifi.click/bureau/v1/tu/report>

Request

```
{
  "includeJsonReport": true,
  "includePdfReport": false,
  "includeSummaryReport": true,
  "clientReference": "test",
  "title": "Mr" ,
  "forename": "Ensom",
  "middleName": "",
  "surname": "AutomotiveFinance",
  "dateOfBirth": "1962-01-01",
  "currentAddress": {
    "flat": "1",
    "street": "Fisher Street",
    "postTown": "TEST TOWN",
    "postCode": "X9 1WS"
  },
}
```

```
"previousAddress": null,  
"previousPreviousAddress": null  
}
```

Response

A simplified sample response is provided below for clarity.

Sample reports will be provided alongside this document.

The response can include three components: the raw credit report in JSON format, a summary report, a PDF version of the raw credit report, a PDF of the summary report.

Please note that including the PDF significantly increases response time, so we recommend retrieving it in a separate request.

```
{  
  "status": "true",  
  "data": {  
    "id": "dea5f3b8-c5f8-4b59-a7ef-3852a4d24674",  
    "jsonReport": {  
      "$": {  
        "xmlns:i": "http://www.w3.org/2001/XMLSchema-instance"  
      },  
      "Message": {  
        "$": {  
          "i:nil": "true"  
        }  
      },  
      "Status": "true",  
      "Report": {},  
      "summaryReport": {  
        "matchedAddress": {  
          "$": {  
            "i:nil": "true"  
          }  
        },  
      },  
    },  
  },  
}
```

```

    "totalAccounts": 12,
    "hpAccounts": 11,
    "leaseAccounts": 1,
    "totalInReport": 12,
    "accounts": [
      {
        "accountNumber": "*****2992 1",
        "lenderName": "Callcredit",
        "name": "Mr Ensom Automotivefinance",
        "address": "1 Fisher Street Test Town X9 1WS",
        "dob": "1962-01-01T00:00:00",
        "startDate": "2010-07-05T00:00:00",
        "endDate": "2014-05-07T00:00:00",
        "accountType": "HP",
        "accountTypeName": "Hire Purchase",
        "monthlyPayment": "500",
        "currentStatus": "Settled",
        "startBalance": null,
        "currentBalance": "0",
        "defaultBalance": null,
        "sourcedFrom": "API"
      }
    ]
  },
  "pdfReport": "Base 64 Data"
}

```

Notes:

- The [data.id](#) should be stored and can be used to retrieve the saved report at a later date.
- The report can return either a single object or a collection of objects when multiple items exist. This affects how you access the data in your integration.

Summary Report

Both the initial request for a report and the subsequent retrieval of a previous report, can include a summary report by passing in includeSummaryReport: true.



In most cases, the summary report will be the preferable data to retrieve and work with

The summary report is highly advantageous as it negates the need to interrogate the raw credit file. The summary report will also include batch data if it has been added (usually 24 hours after the original request)

The summary report can also be retrieved as a PDF.

Example

```
{
  "status": "true",
  "data": {
    "id": "dea5f3b8-c5f8-4b59-a7ef-3852a4d24674",
    "summaryReport": {
      "matchedAddress": {
        "$": {
          "i:nil": "true"
        }
      },
    },
    "totalAccounts": 12,
    "hpAccounts": 11,
    "leaseAccounts": 1,
    "totalInReport": 12,
    "accounts": [
      {
        "accountNumber": "*****2992 1",
        "lenderName": "Callcredit",
        "name": "Mr Ensom Automotivefinance",
        "address": "1 Fisher Street Test Town X9 1WS",
      }
    ]
  }
}
```

```

        "dob": "1962-01-01T00:00:00",
        "startDate": "2010-07-05T00:00:00",
        "endDate": "2014-05-07T00:00:00",
        "accountType": "HP",
        "accountTypeName": "Hire Purchase",
        "monthlyPayment": "500",
        "currentStatus": "Settled",
        "startBalance": null,
        "currentBalance": "0",
        "defaultBalance": null,
        "sourcedFrom": "API"
    },
    {
        "accountNumber": "*****2910 1",
        "lenderName": "Transunion",
        "name": "Mr Ensom Automotivefinance",
        "address": "1 Fisher Street Test Town X9 1WS",
        "dob": "1962-01-01T00:00:00",
        "startDate": "2014-07-31T00:00:00",
        "endDate": "2020-05-18T00:00:00",
        "accountType": "HP",
        "accountTypeName": "Hire Purchase",
        "monthlyPayment": "500",
        "currentStatus": "Settled",
        "startBalance": null,
        "currentBalance": "0",
        "defaultBalance": null,
        "sourcedFrom": "API"
    }
]
}
}
}
}

```



The TransUnion summary response structure differs slightly from the Equifax structure. The key difference is that names and addresses are returned in a flat structure rather than Equifax's structured data. The following keys are omitted:

- `repaymentTerm`
- `paymentFrequency`
- `creditLimit`
- `dateLastDelinquent`
- `dateSettled`
- `defaultDate`
- `companyClass`

Notes

- The report contains a summary that makes it easier to check the total motor finance accounts found across both the initial search and the follow-up historic batch retrieval.
- `totalAccounts`: Total unique accounts across both CZ search and historic batch report
- `hpAccounts`: Number of PCP/HP accounts in the total
- `leaseAccounts`: Number of lease accounts in the total
- `totalInReport`: Number of accounts found in the initial API search
- `totalInBatch`: Number of unique accounts found in the historic batch report after deduplication
- `totalInOriginalBatch`: Number of accounts found in the historic batch report before deduplication
- `accounts`: List of unique accounts found
- The address can include an `addressName` and/or `addressNumber`, which are omitted if undefined

Report Historic Retrieval

To retrieve a previous search as either a PDF, JSON string or summary report, pass the report ID (captured in the initial search) to the following endpoint.



The summary report that can be included in this request will include any batch data if available. It may be advantageous to use this approach rather than interrogating the batch data directly, which would require work to deduplicate accounts against the initial api search.

Get: <https://staging-app.valifi.click/bureau/v1/tu/report/{id}?includePdfReport=true>

The available query parameters are:

```
includePdfReport,  
includePdfSummaryReport,  
includeJsonReport,  
includeSumamryReport
```

Setting these parameters to true determines whether a base64 PDF, JSON report or the summary is included in the response.

Response

A sample response can be provided but has been simplified here for document clarity.

```
{  
  "status": "true",  
  "data": {  
    "id": "dea5f3b8-c5f8-4b59-a7ef-3852a4d24674",  
    "jsonReport": {  
      "$": {  
        "xmlns:i": "http://www.w3.org/2001/XMLSchema-instance"  
      },  
      "Message": {
```

```

    "$": {
      "i:nil": "true"
    }
  },
  "Status": "true",
  "Report": {},
  "summaryReport": {
    "matchedAddress": {
      "$": {
        "i:nil": "true"
      }
    },
  },
  "totalAccounts": 12,
  "hpAccounts": 11,
  "leaseAccounts": 1,
  "totalInReport": 12,
  "accounts": [
    {
      "accountNumber": "*****2992 1",
      "lenderName": "Callcredit",
      "name": "Mr Ensom Automotivefinance",
      "address": "1 Fisher Street Test Town X9 1WS",
      "dob": "1962-01-01T00:00:00",
      "startDate": "2010-07-05T00:00:00",
      "endDate": "2014-05-07T00:00:00",
      "accountType": "HP",
      "accountTypeName": "Hire Purchase",
      "monthlyPayment": "500",
      "currentStatus": "Settled",
      "startBalance": null,
      "currentBalance": "0",
      "defaultBalance": null,
      "sourcedFrom": "API"
    }
  ]
},
"pdfReport": "Base 64 Data"

```

```
}  
}
```

PDF as a stream

The Report and Summary PDF can be retrieved as a stream by passing the report ID (captured in the initial search) to the following endpoint.

Get: <https://staging-app.valifi.click/bureau/v1/tu/report-pdf/{id}>.

Get: <https://staging-app.valifi.click/bureau/v1/tu/report-pdf/{id}>.

Batch Data

API reports by default return 6 years of credit data. This includes accounts that may have been opened more than 6 years ago but remained active within the past 6 years.

Once an API search has been performed, TramsUnion will search their historic database and provide the data to Valifi, typically within 24 hours. This historic data can then be retrieved from Valifi through a simple API call.

Get: <https://api.valifi.co.uk/bureau/v1/tu/batch-timestamps?dateFrom=2025-01-01&dateTo=2025-05-01>

The available query parameters are:

```
dateFrom (required),  
dateTo (required,  
id (optional) ,  
clientRef (optional)
```

A typical process would be to use the date filter to search for all reports within a specific date range. This returns a collection of searches, each containing an ID, your original search reference, and the date it was received back. These dates may be null for recent searches, or batchReceivedAt remains null until the batch is received and processed.

```
{
  "status": true,
  "data": [
    {
      "id": 343534,
      "clientRef": "67d14c8de5cf78963e9dba28",
      "requestAt": "2025-03-12T09:00:56.000Z",
      "batchReceivedAt": "2025-03-13T10:10:13.000Z"
    },
    {
      "id": 343535,
      "clientRef": "67d2acccccdc6c873cc263be",
      "requestAt": "2025-03-13T10:04:31.000Z",
      "batchReceivedAt": null
    }
  ]
}
```

Once your system has a list of IDs with confirmed batch data returned, you can use the history retrieval endpoint to retrieve the summary report. This will give you a clean, deduplicated set of motor finance accounts, highlighting whether each account was found in the original API search or the subsequent batch process.

Get: <https://api.valifi.co.uk/bureau/v1/tu/report/3220?includeSummaryReport=true>

Testing

The Valifi staging environment that you have been set up with is connected to the TransUnion UAT environment.

TransUnion Test Cases

TransUnion provides an extensive list of test customer details that should be included in the request. TransUnion test cases will be provided to you and should cover all possible data scenarios.

Simplified Test Cases

Valifi has simplified test cases down to a few key test customers that cover the most common responses. You can find these test customers in the TransUnion test pack.

To retrieve automotive accounts, use the following details:

```
"forename": "Ensom",  
"surname": "AutomotiveFinance",  
"dateOfBirth": "1962-01-01",  
"currentAddress": {  
  "flat": "1",  
  "street": "Fisher Street",  
  "postTown": "TEST TOWN",  
  "postCode": "X9 1WS"  
},
```

Testing with Postman

A Postman collection is included. Simply add your provided user credentials to the collection variables to instantly interact with the Valifi APIs. Initial test input data has been added to the request, which you can modify using any of the test customers from the test pack.

Testing with the Client Dashboard

By logging into the client interface at <https://www.valifi.click/>, you can manually request reports and AML information. The request page provides a pre-defined list of test customers for quick selection.

Request TU Report

Test Cases

Ensom AutomotiveFinance | 1



Your Reference

ref-25sms6091

Title

Mr



First Name

Ensom

Last Name

AutomotiveFinance

Date of Birth

01/01/1962



At the bottom of the page, you will be able to view the request and response JSON, allowing you to quickly compare with your own code.

Request Data:

```
√ { 11 items
  "includeJsonReport": bool true
  "includePdfReport": bool false
  "clientReference": string "ref-25sms6091"
  "title": string "Mr"
  "forename": string "Ensom"
  "middleName": string ""
  "surname": string "AutomotiveFinance"
  "dateOfBirth": string "1962-01-01"
  > "currentAddress": {...} 10 items
  "previousAddress": null
  "previousPreviousAddress": null
}
```

Response Data:

```
√ { 2 items
  "status": string "true"
  > "data": { 2 items
    "id": string "e45c9381-d0dd-4a98-9e48-f0a6b9f5172d"
    > "jsonReport": { 4 items
```

Interrogating Report

The credit report is a JSON file whose structure varies based on the customer data returned. An official data dictionary will be provided, along with example credit files for the simplified test cases.

A typical report can include:

- **AddressLinks**
- **Cifas**
- **ElectoralRoll**
- **FinancialAccountInformation**
- **FinancialConnections**
- **Moda**
- **NoticesOfCorrection**
- **OtherNames**

- **PersonalInformation**
- **PublicInformation**
- **Rating**
- **ReportDetails**
- **SearchHistory**

It's important to note that if data is not available, the corresponding element will not exist in the report.

If there is only one set of data for a given element, the children of the element will be key-value pairs.

If there are multiple sets of data for a given element, the child of the element will be an array, with its children containing the key-value pairs. This means it is essential to check if the element exists, and then check if it is an array of data. This is a result of converting XML data to JSON format.

Examples:

It's helpful to normalise the data into arrays, ensuring you're always working with an array of data—even when there's only one object in the array.

```
const searchHistory = jsonReport?.Report?.SearchHistory;

if (searchHistory && typeof searchHistory === 'object') {
  console.log('searchHistory found');
  for (const key in searchHistory) {
    const record = searchHistory[key]?.SearchRecord;

    if (record && !Array.isArray(record)) {
      console.log('searchHistory create array');
      searchHistory[key].SearchRecord = [record];
    } else {
      console.log('searchHistory already array');
    }
  }
}
```

Appendix

Mobile ID

Object	Data Type	Description
Phone	Block	The Phone block contains all the information that relates to telephone numbers.
MobileID	Block	Contains information on associations between the telephone number and identity details.
TotalPositiveMatches	Alphanumeric	Total number of positive associations found (InputIdentityMatches.NumMatches + LinkedIdentityMatches.NumMatches + InputAddressMatches.NumMatches).
InputIdentityMatches	Block	Associations between the telephone number and input identity details (name and address).
NumMatches	Alphanumeric	Number of positive associations found.
NumSources	Alphanumeric	Number of unique sources for the matches.
FirstMatchSeen	Date	Date of the first match.
LastMatchSeen	Date	Date of the last match.
LinkedIdentityMatches	Block	Associations between the number and identities linked to the input (e.g. previous address, maiden name).
NumMatches	Alphanumeric	Number of positive associations.
NumSources	Alphanumeric	Number of unique sources.
FirstMatchSeen	Date	Date of first match.
LastMatchSeen	Date	Date of last match.
InputAddressMatches	Block	Associations between the number and the address provided in the search.
NumMatches	Alphanumeric	Number of matches found.

Object	Data Type	Description
NumSources	Alphanumeric	Unique sources of the address match.
FirstMatchSeen	Date	Date of first match.
LastMatchSeen	Date	Date of last match.
UnknownMatches	Block	Associations between the number and identities <i>not</i> linked to the input name and address.
NumMatches	Alphanumeric	Number of matches found.
NumSources	Alphanumeric	Unique sources for those associations.
FirstMatchSeen	Date	Date of first association.
LastMatchSeen	Date	Date of last association.
UnknownMatchDetails	Block	Details on unknown identities associated with the input telephone number.
NumUniqueIdentities	Alphanumeric	Number of unique identities associated.
NumUniqueAddresses	Alphanumeric	Number of unique addresses associated.